

Week 8

Deep Learning

Question: 1

A fintech startup is building a deep neural network to predict credit risk. The model initially performs well, but after increasing its depth, the training loss stops improving. What is the most plausible reason for this?

Options:

- A) Overfitting due to small dataset size
- B) Vanishing gradients due to deep architecture
- C) Underfitting due to low capacity
- D) High learning rate causing exploding gradients

Correct Answer: B

Explanation: Deeper networks often suffer from vanishing gradients, especially when using activations like sigmoid. This leads to poor weight updates in early layers.

Question: 2

What causes gradients to vanish in deep neural networks?

Options:

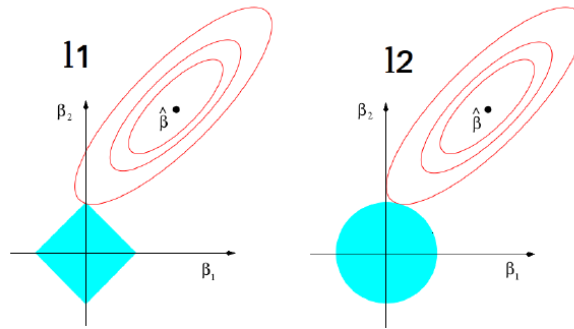
- A) Derivative of sigmoid is small when input is large or small
- B) Chain rule leads to multiplication of several small terms
- C) Weight matrices are sparse
- D) Gradients are clipped to avoid overflow

Correct Answers: A, B

Explanation: Both (A) and (B) lead to vanishing gradients: the small derivative of sigmoid and chain rule multiplication of small numbers.

Question: 3

Consider two models trained on the same dataset: Model A uses L1 regularization and Model B uses L2 regularization. The following figure shows the contour plots of the loss function and the respective constraint regions for both models.



Based on this, which of the following statements are correct?

Options:

- A) Model A is more likely to produce sparse weight vectors (i.e., with many zero values)
- B) L1 regularization tends to force weights toward the origin but does not make them exactly zero
- C) L2 regularization penalizes large weights but allows all weights to stay non-zero
- D) The diamond-shaped constraint region also corresponds to L2 regularization

Correct Answers: A, C

Question: 4

What if a Deep Neural Network uses only linear transformations (i.e., no non-linear activation functions)?

Options:

- A) It can learn highly non-linear and complex decision boundaries
- B) It becomes equivalent to a single-layer linear model
- C) It performs better due to reduced complexity
- D) It requires fewer parameters to train

Correct Answer: B

Explanation: A DNN with only linear transformations collapses into a single linear function, regardless of the number of layers. This limits its expressiveness and prevents it from learning complex, non-linear patterns in the data.

Question: 5

An engineer trains a deep network, but loss doesn't decrease. Gradients are near zero after a few epochs. What is the best action to fix the issue?

Options:

- A) Increase the batch size
- B) Use linear activation in all layers
- C) Use better initialization or replace sigmoid with ReLU

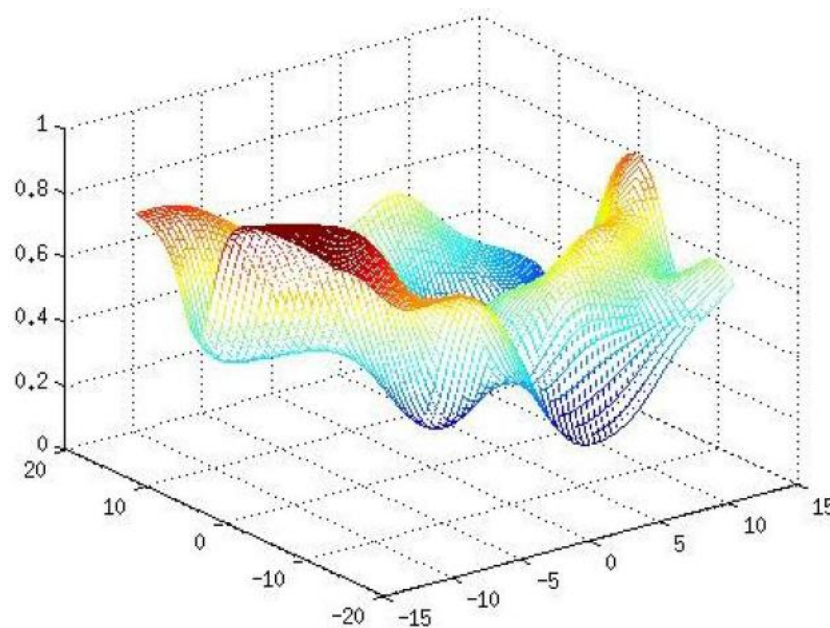
D) Reduce the learning rate

Correct Answer: C

Explanation: ReLU does not saturate like sigmoid and helps prevent vanishing gradients. Good initialization also helps gradients flow better.

Question: 6

The figure shows the error surface of a deep neural network (DNN), which is highly non-convex with multiple valleys and plateaus. A research team is training a small-capacity DNN and finds it frequently stuck in poor local minima. They consider applying unsupervised pre-training to help improve performance.



What does the error surface in the image imply about training deep neural networks?

Options:

- A) The loss surface is convex and easy to optimize
- B) The model has only one global minimum
- C) The optimization landscape contains many local minima and flat regions
- D) Pre-training always guarantees convergence to the global minimum

Correct Answer: C

Explanation: The surface is highly non-convex, indicating many local minima and plateaus, which make optimization difficult.

Data for questions 7 and 8

A team is training a deep network using only supervised learning. They observe that while the model achieves low training error, it often overfits or converges to poor generalization regions. To improve performance, they consider unsupervised pre-training, where the network first

minimizes reconstruction loss (unsupervised objective - $\Omega(\theta)$) before fine-tuning for prediction (supervised objective - $L(\theta)$).

The objective functions are

$$\Omega(\theta) = \frac{1}{m} \sum_{i=1}^m \sum_{j=1}^n (x_{ij} - \hat{x}_{ij})^2$$
$$L(\theta) = \frac{1}{m} \sum_{i=1}^m (y_i - f(x_i))^2$$

Question: 7

What role does the unsupervised objective $\Omega(\theta)$ serve in this setup?

Options:

- A) It reduces the test error directly
- B) It provides a constraint on the parameter space
- C) It optimizes only the last layer of the network
- D) It increases the greediness of supervised learning

Correct Answer: B

Explanation: The unsupervised objective acts as a regularizing constraint on the optimization, ensuring that weights lie in regions that preserve important data structure.

Question: 8

What is a likely benefit of minimizing $\Omega(\theta)$ before $L(\theta)$?

Options:

- A) It guarantees a global minimum in supervised training
- B) It prevents learning features from labeled data
- C) It improves generalization by better parameter initialization
- D) It reduces the number of hidden layers required

Correct Answer: C

Explanation: Unsupervised pre-training guides the optimization into better regions of the parameter space, leading to faster convergence and better generalization in supervised tasks.

Question: 9

An engineer is building a deep network for image classification but struggles to train it from scratch. A colleague suggests training each layer as an autoencoder before fine-tuning the entire network.

Which training method is being used here?

Options:

- A) Supervised learning with batch normalization
- B) Layer-wise greedy unsupervised pre-training
- C) Reinforcement learning
- D) Transfer learning

Correct Answer: B

Explanation: In greedy layer-wise pre-training, each layer is trained as an autoencoder to reconstruct its input before moving to the next layer.

Question: 10

A student is training a deep network for digit recognition using sigmoid activation in all layers. Although the architecture is correct, the network's performance doesn't improve even after many epochs. Upon inspecting the gradients, they appear close to zero in early layers. What is the most likely cause for the near-zero gradients?

Options:

- A) ReLU function is not working
- B) Activation values are saturated at extreme values
- C) Output labels are not one-hot encoded
- D) The optimizer is not configured properly

Correct Answer: B

Explanation: Sigmoid outputs values between 0 and 1. For large or small inputs, its derivative becomes nearly zero, causing vanishing gradients.

Question: 11

Your team is struggling to train a deep network using sigmoid activation. You are asked to suggest alternative activation functions that overcome its limitations. Which of the following activation functions address the problems of vanishing gradients and zero-centering?

Options:

- A) Tanh
- B) Softmax
- C) ReLU
- D) Linear

Correct Answers: A, C

Explanation: Tanh is zero-centered, unlike sigmoid. ReLU avoids saturation (for positive values) and helps mitigate vanishing gradients.

Question: 12

What is the derivative of sigmoid at $x = 10$?

Options:

- A) 1
- B) 0.5
- C) Close to 0
- D) Undefined

Correct Answer: C

Explanation:

$$\sigma(10) \approx 1$$
$$\sigma'(x) = \sigma(x)(1 - \sigma(x)) \approx 0$$

Question: 13

What increases the risk of dying ReLU neurons?

Options:

- A) Negative bias initialization
- B) Large learning rate
- C) Zero-centered inputs
- D) Xavier initialization

Correct Answers: A, B

Explanation: If ReLU receives negative inputs, it outputs zero and may stop updating (i.e., "die"). Bad initialization and large learning rates can cause this.

Question: 14

Select a correct option for the following figures

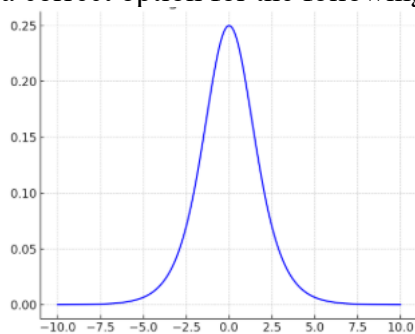


Figure A

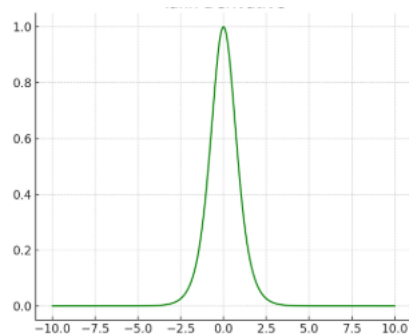


Figure B

Options:

- A) Figure A is derivative of ReLU
- B) Figure B is derivative of ReLU
- C) Figure A is derivative of Sigmoid
- D) Figure B is derivative of Sigmoid
- E) Figure A is derivative of Tanh
- F) Figure B is derivative of Tanh

Correct Answer: C, F

Question: 15

Select the correct match of activation functions with the description.

Activation Function	Description
A. Sigmoid	1. Commonly used in LSTMs/RNNs
B. ReLU	2. Often leads to vanishing gradients
C. Leaky ReLU / Maxout	3. Standard unit in CNNs
D. Tanh	4. Explored as alternative to handle dead units

Options:

- A) A-2, B-3, C-4, D-1
- B) A-1, B-2, C-3, D-4
- C) A-4, B-1, C-2, D-3
- D) A-3, B-4, C-1, D-2

Correct Answer: A

Information for three questions (16 to 18)

A student is experimenting with a neural network using ReLU activations. They observe that after a few epochs, several neurons are not contributing to the output at all. They notice their learning rate is quite high, and all biases are initialized to 0. They are unsure how this impacts learning.

Question: 16

What is the primary reason neurons may "die" when using ReLU activations?

Options:

- A) ReLU is a nonlinear function
- B) ReLU's gradient becomes zero for negative inputs
- C) ReLU saturates in the positive region
- D) Bias is always fixed in ReLU

Correct Answer: B

Explanation: For ReLU, if the input to a neuron is negative, the derivative is 0, so weights stop updating. This can permanently deactivate a neuron, a phenomenon known as the dying ReLU problem.

Question: 17

Which of the following strategies can help prevent the dying ReLU problem?
Select all that apply:

Options:

- A) Using a very large learning rate
- B) Initializing bias to a small positive value
- C) Using sigmoid instead of ReLU
- D) Using variants like Leaky ReLU
- E) Setting weights to zero

Correct Answers: B and D

Explanation: Positive biases (e.g., 0.01) ensure some positive activations early in training. Leaky ReLU and other variants retain small gradients even when input is negative.

Question: 18

Why does ReLU help speed up convergence compared to sigmoid or tanh?

Options:

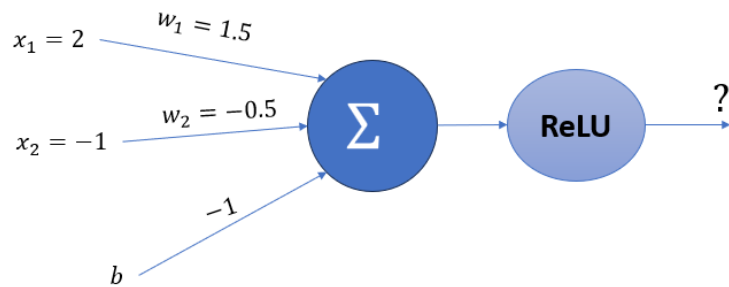
- A) It has a simpler formula
- B) Its gradient never vanishes
- C) It avoids saturation in positive regions
- D) It uses fewer parameters

Correct Answer: C

Explanation: ReLU does not saturate in the positive region, so gradients remain large for positive inputs. This avoids the vanishing gradient problem seen with sigmoid/tanh.

Question: 19

You are given the input to a ReLU neuron as:



Options:

- A) 2.5
- B) 3.0
- C) 2.0
- D) 1.0

Correct Answer: A

Question: 20

For a neuron using ReLU, you observe that its output remains 0 for all training samples, and its pre-activation $z \leq -0.1$ always. What is the effective gradient flowing through it?

Options:

- A) Same as output gradient
- B) 0
- C) Depends on weights
- D) 1

Correct Answer: B

Explanation: Since the input to ReLU is always negative, the gradient is 0, and the neuron is effectively dead.

Question: 21

A neuron has input $x = -5$. Using Leaky ReLU with slope 0.01, what is the output?

Options:

- A) 0
- B) -0.5
- C) -5
- D) -0.05

Correct Answer: D

Explanation: $f(x) = \max(0.01(-5), -5) = -0.05$

Question: 22

Assume a Parametric ReLU with $\alpha = 0.2$. For input $x = -3$, what is the output?

Options:

- A) 0
- B) -0.6
- C) -3
- D) -1.5

Correct Answer: B

Explanation: $f(x) = \max(\alpha x, x) = \max(-0.6, -3) = -0.6$

Question: 23

Which of the following is true about Leaky ReLU compared to standard ReLU?

Options:

- A) Leaky ReLU allows small gradients even for negative inputs
- B) Leaky ReLU avoids dead neuron problem
- C) Leaky ReLU saturates at large positive values
- D) Leaky ReLU is more computationally expensive than Sigmoid

Correct Answers: A, B

Explanation: Leaky ReLU ensures gradients flow for negative values (solving dead neuron issue). It does not saturate, and it's computationally simple.

Question: 24

A data scientist is training a deep neural network to classify customer sentiment. Initially, they used ReLU as the activation function. However, they observed that some neurons were dying during training, especially for inputs near or below zero. They switched to Exponential Linear Unit (ELU) activation with $\alpha = 1$, hoping to retain some gradient flow even for negative input values.

During debugging, the scientist prints the outputs from an intermediate layer for three inputs:

- Input A: $x = -1$
- Input B: $x = 0$
- Input C: $x = 2$

Based on the properties of ELU, which of the following are true?

Options:

- A) Input A will output approximately -0.63
- B) Input B will output 1
- C) Input C will output 2
- D) ELU provides small but non-zero gradient when $x < 0$, preventing dying neurons
- E) ELU is computationally cheaper than ReLU

Correct Answers: A, C, D

Question: 25

Which of the following statements is true about the Maxout Neuron?

Options:

- A) It uses a minimum function over multiple linear combinations of inputs
- B) It always suffers from saturation for negative inputs
- C) It requires more parameters compared to ReLU
- D) It can only approximate sigmoid-like functions

Correct Answer: C

Question: 26

Why is it important to initialize weights randomly in a deep neural network instead of initializing them all to the same value?

Options:

- A) To increase training time
- B) To break symmetry and allow neurons to learn different features
- C) To reduce variance in predictions
- D) To prevent overfitting by regularization

Correct Answer: B

Question: 27

In Batch Normalization, which of the following parameters are learned during training?

Options:

- A) Mean and Variance
- B) Mean and ϵ
- C) γ (scale) and β (shift)
- D) Variance and ϵ

Correct Answer: C

Explanation: BatchNorm normalizes activations using batch statistics (mean & variance — computed, not learned) and then learns γ and β parameters to allow the network to scale and shift the normalized values appropriately.